Cryptanalysis Walkthrough Report

(i) Recovered Plaintext

BREAKER OF THIS CODE WILL BE BLESSED BY THE SQUEAKY SPIRIT RESIDING IN THE HOLE GO AHEAD AND FIND A WAY OF BREAKING THE SPELL ON HIM CAST BY THE EVIL JAFFAR THE SPIRIT OF THE CAVE MAN IS ALWAYS WITH YOU FIND THE MAGIC WAND THAT WILL LET YOU OUT OF THE CAVES IT WOULD MAKE YOU A MAGICIAN NO LESS THAN JAFFAR SPEAK THE PASSWORD THE MAGIC OF WAND TO GO THROUGH

(ii) Encryption Algorithm/Cipher Name

The adversary utilized a two-layered substitution-permutation network (SPN):

- **Transposition Layer:** A columnar transposition cipher was applied using a block size of 5 and a key order of [3, 2, 4, 0, 1].
- **Substitution Layer:** A monoalphabetic substitution cipher was applied to the resulting text, mapping individual characters according to a provided key dictionary.

(iii) Encryption Keys

Permutation Order: [3, 2, 4, 0, 1]

Substitution Key (Inverse Mapping):

```
q:a, j:b, e:c, p:d, v:e, s:f, g:g, f:h, c:i, k:j,
m:k, t:l, u:m, y:n, w:o, h:p, i:q, n:r, l:s, a:t,
d:u, b:v, r:w, o:x, x:y, z:z
```

(iv) Approach for Problem Solution

To recover the original message, the decryption process was performed in the inverse order of encryption:

1. **Data Cleaning:** The ciphertext was sanitized to remove spaces, leaving only the raw encrypted characters.
2. **Reverse Transposition:** A Python function was employed to process the clean ciphertext in blocks of 5. By reordering the characters in each block according to the permutation [3, 2, 4, 0, 1], the text was returned to the state it was in after the substitution phase but before the transposition.
3. **Reverse Substitution:** Using the inverse substitution dictionary, each character in the permuted string was mapped back to its original plaintext equivalent.
4. **Formatting:** The resulting string was converted to uppercase, yielding the final recovered message.